

# Jonathan Vacher

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## Research Experience

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### Associate Professor (Maître de conférences)

MAP5, CNRS, UFR de Mathématiques et Informatique  
Applied Mathematics

Université Paris Cité, Paris, France

Sept. 2022

### Agrégé Teacher in Mathematics

Académie de Versailles, France

On research leave from 2016 to 2022.

### Postdoctoral Research Fellow

Laboratoire des Systèmes Perceptifs, École Normale Supérieure  
Supervisors: Pascal Mamassian and Ruben Coen-Cagli

PSL Research University, Paris, France

Sept. 2020 - Aug. 2022

### Postdoctoral Research Fellow

Department of Computational Biology  
Supervisors: Ruben Coen-Cagli and Pascal Mamassian

Albert Einstein College of Medicine, New York, USA

Sept. 2017 - Aug. 2020

### PhD Student

Ceremade, Dauphine University  
Unité Neurosciences, Information et Complexité (NeuroPsi)  
Gif-sur-Yvette, France

PSL Research University, Paris, France

Oct. 2013 - Aug. 2017

Supervisors: Gabriel Peyré and Cyril Monier

Title: Dynamic Textures Synthesis for Probing Vision in Psychophysics and Electrophysiology

## Research Supervision

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### Current

2026– Arghavan Aslani (M1-M2), Margaret Bowen (M2)

2025– César Denerier (PhD)

### Alumni

2025 Ruikai Chen (M2), Tomasz Kostrzewa (M2), César Denerier (M2)

2024 Félix Watine (M2)

2021 Matteo Dutertre (M1)

2020 Elliot Kim (PhD)

2020 Alexander Ferrena (PhD)

## List of Contributions

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### Preprints

until May 2026

- Vacher, J., « Deep neural networks as iterated function systems and a generalization bound », *arXiv preprint arXiv:2601.19958*, 2026
- Biswas, T. K., Vacher, J., Molholm, S., Mamassian, P., Coen-Cagli, R., « Natural scene segmentation dynamics reveal iterative bayesian inference », *bioRxiv*, pp. 2026–01, 2026

## Journal Articles

until May 2026

- Meso, A. I., **Vacher, J.**, Gekas, N., Mamassian, P., Perrinet, L. U., Masson, G., « Dyntex: a real-time generative model of dynamic naturalistic scenes from luminance textures », *Journal of Vision*, vol. 25, no. 11, pp. 2–2, 2025
- **Vacher, J.**, Launay, C., Mamassian, P., Coen-Cagli, R., « Measuring uncertainty in human visual segmentation », *PLOS Computational Biology*, vol. 19, no. 9, pp. 1–24, Sep. 2023
- **Vacher, J.**, Launay, C., Coen-Cagli, R., « Flexibly regularized mixture models and application to image segmentation », *Neural Networks*, vol. 149, pp. 107–123, 2022
- **Vacher, J.**, Briand, T., « The Portilla-Simoncelli Texture Model: towards Understanding the Early Visual Cortex », *Image Processing On Line*, vol. 11, pp. 170–211, 2021, <https://doi.org/10.5201/ipol.2021.324>
- Le Coënt, A., Fribourg, L., **Vacher, J.**, Wisniewski, R., « Probabilistic reachability and control synthesis for stochastic switched systems using the tamed euler method », *Nonlinear Analysis: Hybrid Systems*, vol. 36, p. 100 860, 2020
- Roggerone, V., **Vacher, J.**, Tarlaoui, C., Guastavino, C., « Auditory motion perception emerges from successive sound localizations integrated over time », *Scientific Reports*, vol. 9, p. 16 437, 2019
- **Vacher, J.**, Meso, A. I., Perrinet, L. U., Peyré, G., « Bayesian modeling of motion perception using dynamical stochastic textures », *Neural computation*, vol. 30, no. 12, pp. 3355–3392, 2018
- Briand, T., **Vacher, J.**, « How to apply a filter defined in the frequency domain by a continuous function », *Image Processing On Line*, vol. 6, pp. 2016–11, 2016
- Briand, T., **Vacher, J.**, Galerne, B., Rabin, J., « The heeger-bergen pyramid-based texture synthesis algorithm », *Image Processing On Line*, vol. 4, pp. 2014–11, 2014

## Conference Proceedings

until May 2026

- Le Coënt, A., **Vacher, J.**, Kergrene, K., « Formal stabilization of a coupled ode-pde switched system », in *2024 IEEE 63rd Conference on Decision and Control (CDC)*, IEEE, 2024, pp. 6341–6348
- **Vacher, J.**, Mamassian, P., « Perceptual scales predicted by fisher information metrics », in *The Twelfth International Conference on Learning Representations*, 2024
- Launay, C., **Vacher, J.**, Coen-Cagli, R., « Unsupervised video segmentation algorithms based on flexibly regularized mixture models », in *2022 IEEE International Conference on Image Processing (ICIP)*, IEEE, 2022, pp. 4073–4077
- **Vacher, J.**, Davila, A., Kohn, A., Coen-Cagli, R., « Texture interpolation for probing visual perception », 2020
- Le Coënt, A., Fribourg, L., **Vacher, J.**, « Control synthesis for stochastic switched systems using the tamed euler method », in *6th IFAC Conference on Analysis and Design of Hybrid Systems ADHS 2018*, vol. 51, 2018, pp. 259–264
- **Vacher, J.**, Meso, A. I., Perrinet, L. U., Peyré, G., « Biologically inspired dynamic textures for probing motion perception », in *Advances in Neural Information Processing Systems*, 2015

## Unpublished Reports

until May 2026

- **Vacher, J.**, Mamassian, P., Coen-Cagli, R., « Probabilistic model of visual segmentation », *arXiv preprint arXiv:1806.00111*, 2019

## Participation in Conferences and Seminars

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### Upcoming

- Jun. 2026 *Séminaire LMA, CNRS & Aix-Marseille Université, [Link](#)*  
Jun. 2026 *Curves and Surfaces Conference, Saint-Malo, [Link](#)*

### Past

- Mar. 2025 *Séminaire de l'équipe statistiques du MAP5, UPCité, [Link](#)*  
Mar. 2025 *Workshop on Computational Neuroscience, UPCité, [Link](#)*  
Mar. 2025 *Séminaire de modélisation mathématique en sciences de la vie et santé, Paris, [Link](#)*  
Nov. 2024 *Séminaire LMBA, Université Bretagne Sud, [Link](#)*  
May 2024 *The Twelfth International Conference on Learning Representations, Vienna, [Link](#)*  
May 2024 *INC Days, Université Paris Cité*  
Feb. 2024 *Seminar, Institut de la Vision, Paris, [Link](#)*  
Oct. 2023 *Seminar, Imaging in Paris, [Link](#)*  
Apr. 2023 *Journées ANR Mistic, Paris, [Link](#)*  
Jan. 2023 *GDR Vision, Toulouse, [Link](#)*  
Oct. 2022 *Seminar, Imaging in Paris, [Link](#)*  
Mar. 2022 *Seminar, Institut des Neurosciences de la Timone, Marseille, [Link](#)*  
Feb. 2022 *Séminaire de l'équipe Image, Centre Borelli, ENS Paris-Saclay, [Link](#)*  
Dec. 2021 *Séminaire de l'équipe MLMDA, Centre Borelli, ENS Paris-Saclay, [Link](#)*  
Nov. 2021 *Séminaire de l'équipe Parietal, INRIA Saclay, [Link](#)*  
Oct. 2021 *GDR Vision, Lille, [Link](#)*  
Oct. 2021 *NeuroSpin Conferences, Paris-Saclay, [Link](#)*  
Sep. 2021 *Séminaire IMAGES team, Télécom Paris, [Link](#)*  
Aug. 2021 *Modélisation Aléatoire et Statistique (SMAI): Statistique et Image, [Link](#)*  
Jun. 2021 *SMAI Congress: Transport optimal pour l'inférence statistique, France, [Link](#)*  
Jun. 2021 *Séminaire de l'ANR Mistic, [Link](#)*  
May 2021 *Séminaire Images Optimisation et Probabilités de l'IMB, Bordeaux, [Link](#)*  
Dec. 2020 *Spotlight – Neural Information Processing Systems (NeurIPS), online, [Link](#)*  
Dec. 2020 *Vision team seminar at INCC, Université Paris Cité, [Link](#)*  
Nov. 2020 *Image team seminar at MAP5, Université Paris Cité, [Link](#)*  
Nov. 2020 *GDR Vision, online, [Link](#)*  
Jun. 2020 *Vision Science Society (VSS), online, [Link](#)*  
Feb. 2020 *Computational and Systems Neuroscience (Cosyne), Denver, [Link](#)*  
Aug. 2019 *European Conference on Visual Perception (ECVP), [Link](#)*  
Jun. 2016 *International Conference on Mathematical NeuroScience (ICMNS), [Link](#)*  
Dec. 2015 *Spotlight – Neural Information Processing Systems (NeurIPS), Montréal, [Link](#)*  
Jun. 2015 *SMAI Congress, France, [Link](#)*  
Jun. 2015 *International Conference on Mathematical NeuroScience (ICMNS), Nice, [Link](#)*  
Oct. 2014 *Workshop on Geometrical Models in Vision, Paris, [Link](#)*

## Review Activity and Scientific Service

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### Conferences

- 2020–2026 *International Conference on Machine Learning (Area Chair 2025/2026, [Link](#))*  
2020–2026 *International Conference on Learning Representations, [Link](#)*  
2019–2026 *Neural Information Processing Systems (Area Chair since 2024, [Link](#))*

## Journals

|           |                                                                                          |
|-----------|------------------------------------------------------------------------------------------|
| 2019–2023 | <i>Vision Research</i> ( <a href="#">Link</a> )                                          |
| 2019–2023 | <i>Computer Vision and Image Understanding</i> ( <a href="#">Link</a> )                  |
| 2019–2023 | <i>IEEE Transactions on Visualization and Computer Graphics</i> ( <a href="#">Link</a> ) |

## Professional Society Membership

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|           |                                                                                      |
|-----------|--------------------------------------------------------------------------------------|
| 2015–     | <i>Société de Mathématiques Appliquées et Industrielles</i> ( <a href="#">Link</a> ) |
| 2019–2023 | <i>Vision Science Society</i> ( <a href="#">Link</a> )                               |

## Awards and Scholarships

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|           |                                                                                                                                                                                                                                                     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024–2025 | <b>KCL/UPCité Joint Call</b> Université Paris Cité / King’s College London, <b>10 k€</b> ( <a href="#">Link</a> )                                                                                                                                   |
| 2024–2026 | <b>Emergence</b> Université Paris Cité, <b>15 k€</b> ( <a href="#">Link</a> )                                                                                                                                                                       |
| 2024      | <b>Bourse M2 – INC</b> Université Paris Cité, <b>3 k€</b> ( <a href="#">Link</a> )                                                                                                                                                                  |
| Mar. 2023 | <b>Seal of Excellence</b> for the project <i>Decompose the hierarchical process of human visual segmentation</i> , certificate delivered by the <b>European Commission</b> as the institution managing <i>Horizon 2020</i> ( <a href="#">Link</a> ) |
| 2012–2013 | <b>Master Scholarship</b> from the Jacques Hadamard Foundation, <b>10 k€</b> ( <a href="#">Link</a> )                                                                                                                                               |

## Technical Skills

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### Operating Systems

|         |                               |
|---------|-------------------------------|
| Linux   | <i>Regular user and admin</i> |
| Windows | <i>Regular user</i>           |

### Languages

|            |                                                           |
|------------|-----------------------------------------------------------|
| Python     | <i>numpy, scipy, matplotlib, scikit-learn, pytorch</i>    |
| Matlab     | <i>+psychtoolbox</i>                                      |
| JavaScript | <i>basic knowledge (jspsych for online psychophysics)</i> |
| C/C++      | <i>basic knowledge</i>                                    |

### Data Knowledge

|                  |                                                                  |
|------------------|------------------------------------------------------------------|
| Large-scale data | <i>natural images, natural textures</i>                          |
| Psychophysics    | <i>behavioral data</i>                                           |
| Neuroscience     | <i>extracellular recordings, brain optical imaging, EEG, ...</i> |

### Open Source Code

|                       |                                                                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| vseg package          | <i>Reconstruction of probabilistic image segmentation maps from psychophysical measurements in human participants</i> ( <a href="#">Link</a> ) |
| Dynamic Textures      | <i>Motion Clouds, Drifting Gratings</i> ( <a href="#">Link</a> )                                                                               |
| Texture Interpolation | <i>Optimal transport of deep neural network activation distributions</i> ( <a href="#">Link</a> )                                              |
| Portilla-Simoncelli   | <i>Online demo and code (see publications):</i> <a href="#">Link</a>                                                                           |
| Heeger-Bergen         | <i>Online demo and code (see publications):</i> <a href="#">Link</a>                                                                           |
| Miscellaneous         | <a href="#">Link</a>                                                                                                                           |

## Teaching Activities

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|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Since 2024 | <b>Co-director</b> of the CNAI track of the international CogSUP master's program, Link.                                                                           |
| Since 2022 | <b>Associate professor</b> (undergraduate mathematics, CogSUP master's program, MMA master's program. Total: 192 h/year)                                           |
| 2021–2026  | <b>Examiner</b> in mathematics. Oral exam training (2 h/week). High-level students. Classes Préparatoires aux Grandes Écoles, Louis-le-Grand high school, Paris.   |
| 2013–2015  | <b>Lecturer</b> in analysis, linear algebra, and differential calculus (64 h/year). Mid- to high-level students. Dauphine University, Paris.                       |
| Jul. 2013  | Highest French competitive exam in mathematics for academic teaching: <b>Agrégation de mathématiques</b> .                                                         |
| 2011–2013  | <b>Examiner</b> in mathematics. Oral exam training (2 h/week). High-level students. Classes Préparatoires aux Grandes Écoles, Janson de Sailly high school, Paris. |

## Academic Background

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|-----------|----------------------------------------------------------------------------------------------------------------------------------------|
| 2013–2017 | <b>PhD in Applied Mathematics</b> . Dauphine University, PSL Research University.                                                      |
| 2011–2013 | <b>Master's degree in mathematics</b> . Mathematics, vision, and learning. Graduated with honours. École Normale Supérieure de Cachan. |
| 2010–2011 | <b>Bachelor's degree in mathematics</b> . Applied mathematics. Graduated with honours. École Normale Supérieure de Cachan.             |

## Language Proficiencies

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|---------|---------------|
| French  | <i>Native</i> |
| English | <i>Fluent</i> |
| Spanish | <i>Basic</i>  |

## Interests and Other Activities

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|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sports        | rock climbing/bouldering, biking, hiking, tennis, table tennis, handball                                                                                   |
| Video games   | Teamfight Tactics (League of Legends), FPS, RTS, RPG                                                                                                       |
| Board games   | Dune, Terraforming Mars, Megawatt, Smallworld, 7 Wonders, Terra Mystica, Kingdomino, ...                                                                   |
| Miscellaneous | music, cinema, science, politics, ...                                                                                                                      |
| Volunteering  | President of the "Les sENS de l'Art" association in 2012, in charge of organizing the annual arts and music festival of ENS Cachan (budget: 40,000 euros). |